

OpenWebSearch.eu as a Data Source for AI

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Web Search & Web Data - a Critical Infrastructure

- A **critical infrastructure** for society — and now for AI — comparable to satellite navigation
- A **market oligopoly**: controlled by a handful of gatekeepers (Google, Microsoft, Baidu, ...)

Consequences

- **Reduced user choice**
- **Dependence in the information space**
- **Smaller researches and innovators are locked out**



How to enable researchers, innovators and smaller companies to utilise the web as **data resource** for training and fine-tuning AI systems and to develop **AI-based Search Systems** without massive investments in infrastructure?

OpenWebSearch.eu

Building an **Open Index of the Web** to lower the cost for researchers, innovators, and companies to **tap the Web as resource**

Four objectives

1. **Open technology stack** — open pipelines for crawling, preprocessing, indexing
2. **Resource provision** — a network of providers on EuroHPC / AI Factories
3. **Added-value services** — dashboard, tools, data-access APIs
4. **Bootstrapping the ecosystem** — third-party calls, community, use-cases in **search, analytics, and AI**.

14 core partners

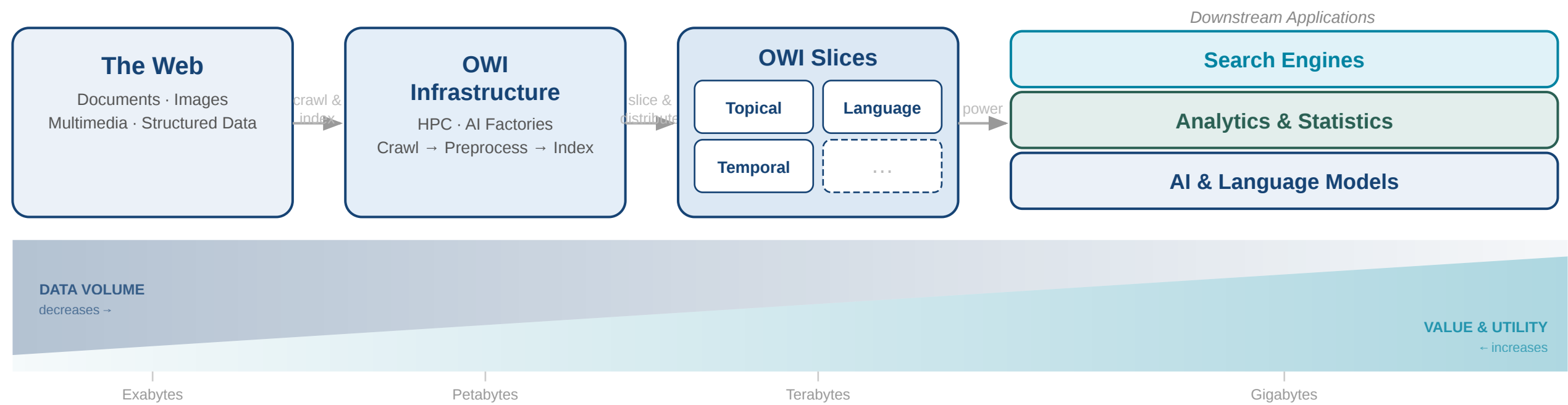


Uni Passau · LRZ · CERN · CSC · IT4I · DLR · TU Graz · Bauhaus-Uni Weimar · Radboud · Uni Leipzig · A1 Telekom · Open Search Foundation · NLnet · SUMA-EV — plus 16 funded third-party projects.

The Goal: An Open Web Index

A **web index** is the data structure behind every search engine — fast, query-based access and ranking over web documents.

Our proposition: a collaboratively created, **open and transparent** web index, running on Europe's HPC centres and AI Factories.



Building the Index

Federated HPC Workflow

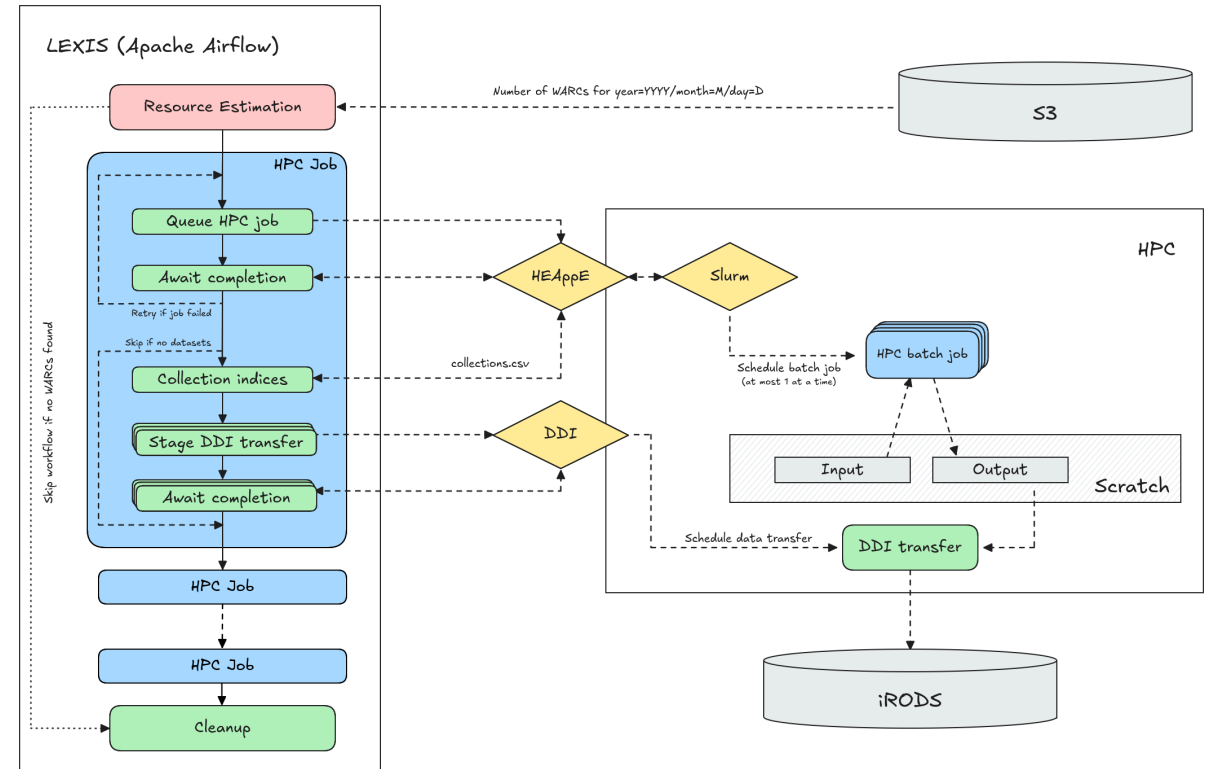
- Continuous Crawling on Cloud Partitions
- Daily Batch HPC Workflows: WARC processing · indexing · semantic enrichment, running
- Runs federated at **LRZ** (Garching) · **IT4I** (Ostrava) · **CSC** (Helsinki)

Federated storage

- Object Storage for Crawl Data
- File Storage for Index Provision
- SQL and slow full-text search for file storage based slice and dice

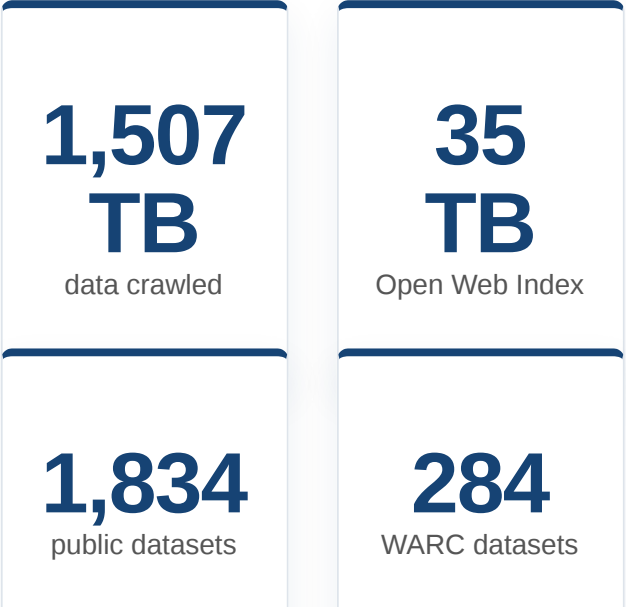
Orchestration

- LEXIS (Apache Airflow) orchestrates jobs across schedulers
- stages data transfers, aligns daily output across all centers



LEXIS / Apache Airflow: federated OWI pipeline orchestration

The Open Web Index Today



Continuously refreshed —
+79 TB crawled and +95 datasets in the last month alone.

OUR²S Web Search Engine



2 Months of data

04/25 + 05/25

598.1M Pages

453.4M Entities

14 verticals

documents · media · products · events · places · FAQs · Scholarly · Artefacts · Actions · Pages · Sites · Persons · Conversations · Courses

in-browser AI

private Qwen3 summaries, ~400 MB



OUR²S turns the OWI into a **working search engine** — documents *and* structured entities, summarised privately in the browser.

OUR²S (ourrs) live demo; search index snapshot April/May 2025.

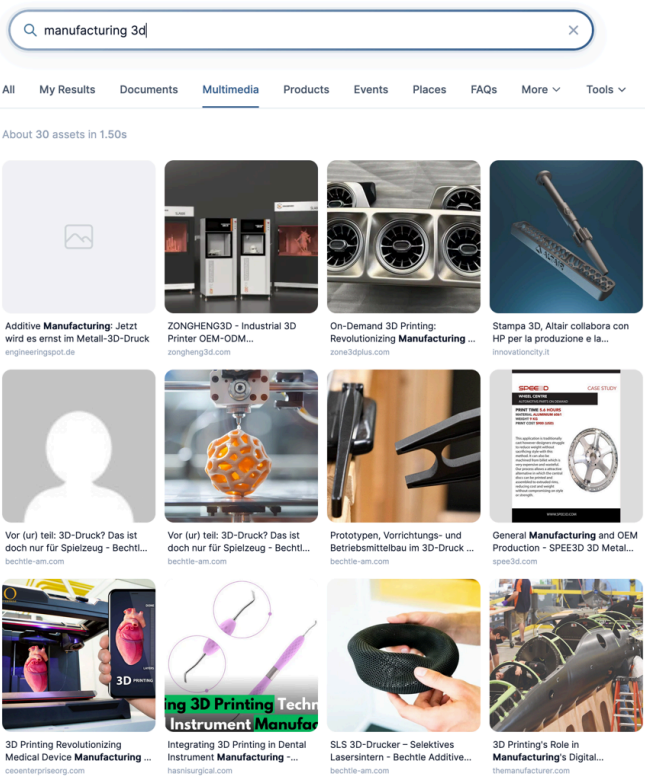
Web data for training & fine-tuning

Modalities: The Index as a Multimodal Filter

MEDIA ASSETS IN ONE MONTH OF THE OPEN WEB INDEX



The index lets you **select training data** by modality, language, quality, and host — *before* paying to download and process it. Storage costs drop when re-downloading text & multimedia can be **targeted and selective**.



Example: multimedia results for a manufacturing query — images, 3D models, product shots.

Domain-Specific Data

"MANUFACTURING" DOCUMENTS IN A SINGLE MONTH OF THE OPEN WEB INDEX

6.3M

"manufacturing"

2.5M

"CAD"

996k

"spare parts"

583k

"3D printing"

498k

"datasheet"

Also present: CNC machining (150k) · injection molding (120k) · industrial automation (562k) · robotics (788k) · PLC (782k) — a domain corpus you can filter out of the OWI.

Keyword matches over one month, English-leaning — a **candidate pool** before curation, dedup, and licensing.

Manufacturing is **just one example** — the OWI holds rich, filterable corpora for *any* domain (law, health, science, finance, engineering, ...), **millions of documents per month**.



Structured Web Data

OWI parses **schema.org / Microdata** and classifies pages — including **forums and Q&A** — into **453.4M typed entities**, with three kinds of value:

~83M Deeper knowledge

articles (79.4M) · scholarly (897k) · books (1.7M) · datasets
· courses · medical
→ **training & RAG**

~39M Behavioural

blog posts (18.2M) · questions (17.5M) · Q&A pages (717k)
· reviews · social posts · forums
→ **instruction & evaluation**

~120M Search & agents

products (91M) · persons (10M) · organisations (6.2M) ·
events (5.2M) · search actions (3.7M)
→ **agentic search & affordances**

The same crawl yields **knowledge to learn from, behaviour to learn from, and entities to act on.**

Search becomes the memory layer

What lives in the weights, and what stays retrievable?

Search as Memory: Parametric vs. Non-Parametric

Retrieval-Augmented Generation (RAG)

Search for context relevant documents.
Usually as Vector search, to cover linguistic and semantic diversity.

RAG introduces a "memory" hierarchy:

- **parametric:** model capabilities & internal knowledge
- **non-parametric:** external knowledge



RAG is essential for freshness, domain specific adaption and data privacy, but requires scalable and efficient **search infrastructures** in an AI system

Web-Scale Vector Memory Has a Cost

The Open Web Index provides **embeddings** for retrieval on selected web documents, but increases compute and storage significantly.

Why compression matters

- Compute is manageable / scalable
- But embeddings are **expensive to store and serve** ($O(n)$ for exact queries)
- compression trades index size against retrieval quality
- Quantisation wins: results hold recall while compressing **16–32×**

Compression essential for **scaling up search**.



From retrieval to agents

Retrieval moves into a control loop — and needs actions and traces.

Agentic Search: Retrieval in a Control Loop



Agents do not just **consume** search results — they **control** retrieval: plan, inspect evidence, call tools, reformulate, synthesize with provenance.

Related paradigms: ReAct (arXiv:2210.03629); Toolformer (arXiv:2302.04761); Self-RAG — adaptive, self-reflective retrieval (arXiv:2310.11511).

A System View of Agentic Search



How much knowledge must sit in **weights** vs. **external memory**? Externalising it → **smaller model + fresher, controllable knowledge** — trading latency & more loops for

Memory roles after CoALA: working · episodic · semantic · procedural (Sumers et al. 2023, arXiv:2309.02427).

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Structured Web Data Becomes Agent Affordance

ACTION & APPLICATION ENTITIES IN THE OPEN WEB INDEX

3.65M

SearchAction

909k

SoftwareApplication

114k

WebApplication

94k

MobileApplication

Agents need **discoverable actions** — what can be searched, called, installed, or configured. The index is a **candidate generator**, not blind trust.

The screenshot shows a search engine interface with the query 'haascnc'. Below the search bar, there are several search results. A red circle highlights the search bar area, which contains the text 'Ergebnisse von haascnc.com' and a search icon. Below the search bar, there are several search results, including 'Haas Automation', 'CNC-Werkzeugmaschinen', 'Product Price List', 'Lathes', and 'Service Quick Links'. A red double-headed arrow points to the search bar area.

haascnc

Ergebnisse von haascnc.com

Haas Automation
https://www.haascnc.com · Diese Seite übersetzen

Haas Automation Inc. - CNC Machine Tools
Haas Automation is the largest machine tool builder in the western world, manufacturing a complete line of CNC vertical machining centers, ...

CNC-Werkzeugmaschinen
Es können keine weiteren obligatorischen Kosten zur ...

Product Price List
The Haas TL Series Toolroom Lathes are affordable, easy to ...

Lathes
Haas CNC Lathe Overview. Find which Lathe is right for you ...

Service Quick Links
Select Your Service Topic: · Operator's Manuals · How-To ...

*Sitelinks searchbox = schema.org **SearchAction**: agents can call a site's own search. Source: metayer.de*

Entity counts by schema.org primary_type, Open Web Index API (api.ourrs), corpus 2026-05.

Beyond Web Data: Traces & Evaluation

OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

Synthetically creating agent traces

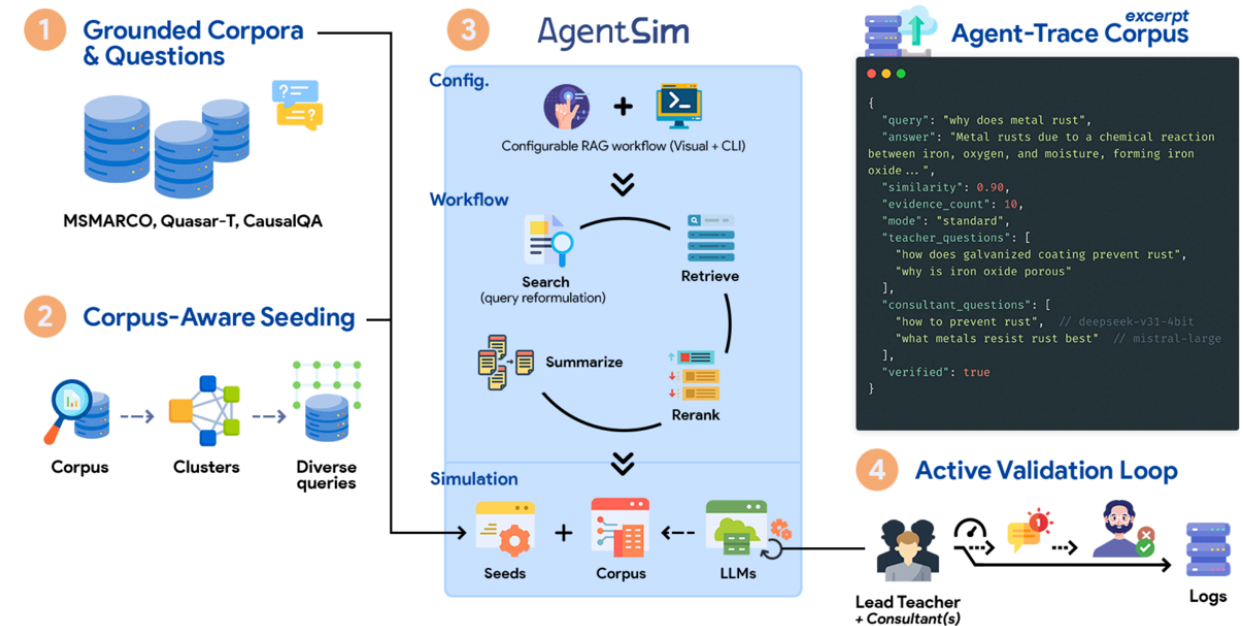
AgentSim — simulates RAG agents over document collections to generate synthetic, **document-grounded** reasoning traces

- **103k+** verifiable steps (Agent-Trace Corpus)
- 100% grounding rate on substantive answers.
- Agent-trace synthetic data generating is both **GPU compute heavy** and requires **search systems**.

AgentSim, Agent-Trace Corpus · arXiv:2604.26653

Comparing to simulated human users

UXSim — grounds an adaptive LLM agent with classical simulators → accurate, **explainable** user-search simulation without costly user studies.



Beyond Web Data: Knowledge & Interaction

OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

How to structure knowledge so agents find it?

NuggetIndex — distil documents into atomic "nuggets": 887 → 320 tokens, far less noise, facts kept for grounding.

Own work — nugget-level indexing for agent knowledge



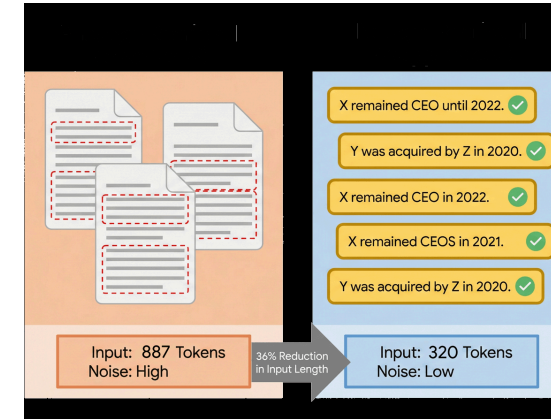
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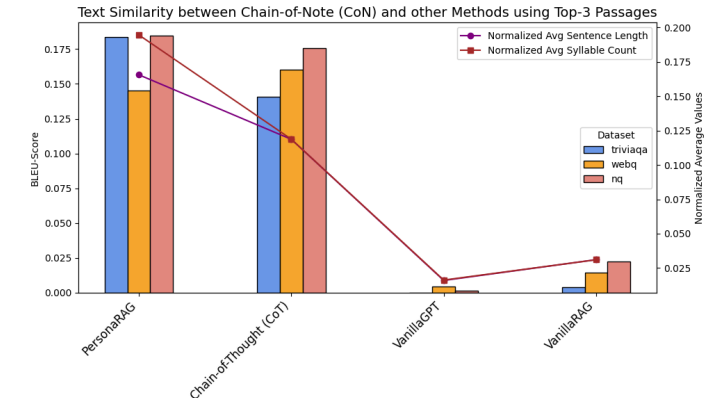
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How should retrieval adapt to the user?

PersonaRAG — user-centric agents personalise retrieval → higher answer quality (BLEU) than CoT and vanilla RAG across TriviaQA, WebQ, NQ.

Zerhoudi & Granitzer 2024 · arXiv:2407.09394



Conclusion



What we covered

- The web is an open, reusable **data layer** — OpenWebSearch delivers **data + an index**
- **Filter** it for training & fine-tuning; **serve** it as RAG memory & agentic search
- Agentic search needs more than data: **interaction, knowledge structure, behaviour, evaluation**
- Developing Agentic Search is not only GPU-intensive, it requires complex systems.

Let's cooperate

We are **looking forward to collaborations** on open web data, retrieval memory, and agentic search — research projects, datasets, and use-cases all welcome.

→ openwebsearch.eu · Chair of Data Science, University of Passau

Better models and better data matter **only when they become systems people use.**

References

OpenWebSearch.eu: <https://openwebsearch.eu/>

TPC26: <https://tpc26.org/> · LRZ goes TPC26: <https://www.lrz.de/en/latest-news/events/detail/lrz-goes-tpc26>

Lewis et al. 2020, Retrieval-Augmented Generation: <https://arxiv.org/abs/2005.11401>

CoRECT 2025, Controlled Retrieval Evaluation of Compression Techniques: <https://arxiv.org/abs/2510.19340>

Kandpal et al. 2023, LLMs Struggle to Learn Long-Tail Knowledge: <https://arxiv.org/abs/2211.08411>

Mallen et al. 2023, When Not to Trust Language Models: <https://arxiv.org/abs/2212.10511>

Yao et al. 2022, ReAct: <https://arxiv.org/abs/2210.03629>

Schick et al. 2023, Toolformer: <https://arxiv.org/abs/2302.04761>

Asai et al. 2023, Self-RAG: <https://arxiv.org/abs/2310.11511>

Sumers et al. 2023, Cognitive Architectures for Language Agents (CoALA): <https://arxiv.org/abs/2309.02427>

Zerhoudi & Granitzer 2024, PersonaRAG: <https://arxiv.org/abs/2407.09394>

AgentSim 2026, Agent-Trace Corpus: <https://arxiv.org/abs/2604.26653>

Zerhoudi & Granitzer 2025, UXSim — Hybrid User Search Simulation: <https://arxiv.org/abs/2602.24241>